Maths for Computing — Phase 0: The Foundations

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How to use these notes

This phase teaches the basic maths tools that every later phase uses. If you skip it, the later topics will feel confusing; if you learn it, they will feel easy.

Your exam covers Lectures 1 to 4 of 4COSC002W Mathematics for Computing. It is multiple choice and closed book, which means no notes and no graphing calculator. So you must be able to do everything with a pen and paper.

How to study each phase:

1. Read every section slowly. Every new word is explained the first time it appears, in **bold**.
2. Work through each worked example with a pen. Do not just read it; copy it.
3. Try the mini tasks at the end of each section **before** looking at the answers.
4. Do the practice questions at the end of the phase. All answers are in the last section, with full explanations.

The full plan (one file per phase):

Phase	Topic	Comes from
0	Foundations: arithmetic, negatives, fractions, powers, algebra, quadratics, graphs, symbols	Background you need
1	Number systems (binary), types of numbers, primes, division theorem, modular arithmetic, ciphers, sequences	Lecture 1, Seminar 1
2	Sets, cardinality, subsets, set builder notation, set operations, Venn diagrams, power sets, Cartesian products, intervals	Lecture 2, Seminar 2
3	Algorithms (search and sort), relations, functions, straight lines and slope, inverses, intersections, recursive functions	Lecture 3, Seminar 3

Phase	Topic	Comes from
4	Logic: propositions, truth tables, implication, valid and satisfiable formulas, logical consequence	Lecture 4
5	Your mock test fully solved, plus a harder practice exam with answers	Mock + revision tasks

When you finish this file, ask for Phase 1.

1. Numbers and the words we use for them

Digits and numbers

A **digit** is a single symbol used to write numbers. In everyday maths there are ten digits: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9. A **number** is built from one or more digits, like 7 or 3471.

The **place value** of a digit is how much it is worth because of where it sits. In 3471, the 3 is worth 3000 because it sits in the thousands place. You will need this idea in Phase 1 for binary.

Kinds of numbers (quick version)

- **Whole numbers**: 0, 1, 2, 3, … (no fractions, no negatives).
- **Negative numbers**: numbers below zero, written with a minus sign, like −3. Think of temperatures below zero.
- **Integers**: all whole numbers and their negatives: …, −3, −2, −1, 0, 1, 2, 3, …
- **Fractions and decimals**: numbers between the integers, like ½ = 0.5.

Phase 1 covers these properly with their official symbols ($\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$).

The four operations and their result words

An **operation** is something you do to numbers. Each one has a name for its answer.

Operation	Symbol	Name of the answer	Example
Addition	+	**sum**	3 + 4 = 7, so the sum is 7

Operation	Symbol	Name of the answer	Example
Subtraction	−	**difference**	9 − 4 = 5, so the difference is 5
Multiplication	× or · or nothing (2x means 2 × x)	**product**	3 × 4 = 12, so the product is 12
Division	÷ or / or a fraction line	**quotient**	12 ÷ 4 = 3, so the quotient is 3

Remainder: the most important word for this module

When a division does not come out exactly, what is left over is the **remainder**.

Example: share 17 sweets between 3 people. Each person gets 5 (because 3 × 5 = 15). There are 17 − 15 = 2 left over. So 17 ÷ 3 = 5 **remainder** 2.

- The number being divided (17) is the **dividend**.
- The number you divide by (3) is the **divisor**.
- The whole-number answer (5) is the **quotient**.
- The leftover (2) is the **remainder**. It is always smaller than the divisor.

How to find a remainder by hand: find the biggest multiple of the divisor that does not go over the dividend, then subtract.

Example: 50 ÷ 6. Multiples of 6: 6, 12, 18, 24, 30, 36, 42, 48, 54. The biggest one not over 50 is 48 = 6 × 8. Remainder = 50 − 48 = 2. So 50 ÷ 6 = 8 remainder 2.

A **multiple** of a number is that number times a whole number. Multiples of 4 are 4, 8, 12, 16, ...

We say a number **divides** another, or the second is **divisible** by the first, when the remainder is 0. 12 is divisible by 4 because 12 ÷ 4 = 3 remainder 0.

Rules for negative numbers

You will use negatives constantly in modular arithmetic, slopes and lines. Learn these rules.

Adding and subtracting — picture a number line. Adding moves right, subtracting moves left.

- 3 − 5 = −2 (start at 3, move 5 left).
- −4 + 6 = 2 (start at −4, move 6 right).
- −2 − 3 = −5 (start at −2, move 3 more left).
- Subtracting a negative is the same as adding: 5 − (−3) = 5 + 3 = 8.
- Adding a negative is the same as subtracting: 5 + (−3) = 5 − 3 = 2.

Multiplying and dividing — look at the signs:

Signs	Result	Example
positive × positive	positive	3 × 4 = 12
negative × negative	positive	(−3) × (−4) = 12
positive × negative	negative	3 × (−4) = −12
negative × positive	negative	(−3) × 4 = −12

Short rule: **same signs give plus, different signs give minus.** The same rule works for division: (−12) ÷ (−4) = 3, and (−12) ÷ 4 = −3.

Absolute value

The **absolute value** of a number is its distance from zero, ignoring the sign. It is written with straight bars: |−7| = 7 and |7| = 7.

Watch out: in Phase 2 the same bars around a **set** mean something different (how many elements it has).

Mini tasks 1

1. Name the answer word: in 20 ÷ 6, what is the quotient and what is the remainder?
2. What is 23 ÷ 7 written as quotient and remainder?
3. Work out: −3 − 8, −5 + 12, 4 − (−6), (−6) × (−7), (−20) ÷ 5.
4. Is 945 divisible by 5? Is 945 divisible by 2?
5. What is |−15|?

2. Order of operations (BIDMAS)

When a calculation has several operations, you must do them in a fixed order, or you get the wrong answer. The order is called **BIDMAS**:

1. **B**rackets — anything inside ( ) first.
2. **I**ndices — powers like 3^2 (covered in section 4).
3. **D**ivision and **M**ultiplication — equal rank, done left to right.
4. **A**ddition and **S**ubtraction — equal rank, done left to right.

Worked examples

Example 1: $2 + 3 \times 4$

- Multiplication before addition: $3 \times 4 = 12$.
- Then $2 + 12 =$ **14**. (Not 20!)

Example 2: $(2 + 3) \times 4$

- Brackets first: $2 + 3 = 5$.
- Then $5 \times 4 =$ **20**.

Example 3: $3^2 - 2 \times 5 + 1$

- Indices: $3^2 = 9$. Now $9 - 2 \times 5 + 1$.
- Multiplication: $2 \times 5 = 10$. Now $9 - 10 + 1$.
- Left to right: $9 - 10 = -1$, then $-1 + 1 =$ **0**.

Example 4 (a recursive formula, like your mock Q19): $3 \times 9 - 2 + 4^2$

- Indices: $4^2 = 16$.
- Multiplication: $3 \times 9 = 27$.
- Left to right: $27 - 2 + 16 =$ **41**.

Brackets inside brackets

Work from the innermost bracket outwards. Example: $2 \times (3 + (4 - 1)) = 2 \times (3 + 3) = 2 \times 6 = 12$.

A minus sign in front of a bracket

A minus in front of a bracket flips every sign inside when you remove the bracket: $-(x - 3) = -x + 3$.

The same idea appears in logic in Phase 4, where operators also have an order of precedence. **Precedence** just means "which goes first".

Mini tasks 2

1. $10 - 4 \div 2$
2. $(10 - 4) \div 2$
3. 2×3^2
4. $(2 \times 3)^2$
5. $5 - (2 - 7)$

3. Fractions and decimals

Fractions appear everywhere in this module: slopes like 9/5, functions like $h(x) = \frac{1}{2}x - \frac{3}{4}$, and inverses like $(x - 3)/2$.

What a fraction is

A **fraction** a/b means "a divided by b". The top number a is the **numerator**. The bottom number b is the **denominator**. So ¾ means 3 ÷ 4 = 0.75.

The denominator can never be 0, because dividing by zero has no meaning. You will see "$b \neq 0$" in rules for this reason. The symbol ≠ means "is not equal to".

Fractions to decimals

Divide the top by the bottom. Learn these by heart:

Fraction	Decimal
1/2	0.5
1/4	0.25
3/4	0.75
1/3	0.333...
2/3	0.666...
1/5	0.2

Fraction	Decimal
10/3	3.333...

Simplifying

Divide the top and bottom by the same number. 6/8 = 3/4 (both divided by 2). 16/4 = 4.

Multiplying fractions

Multiply tops together and bottoms together: (2/3) × (4/5) = 8/15.

A whole number times a fraction: 3 × (1/2) = 3/2 = 1.5. Think of 3 as 3/1.

Example (from your revision sheet): g(x) = (3/4)x − 5/2 at x = −2.

- (3/4) × (−2) = −6/4 = −3/2 = −1.5.
- −1.5 − 2.5 = **−4**.

Dividing fractions

To divide by a fraction, flip it and multiply. (1/2) ÷ (3/4) = (1/2) × (4/3) = 4/6 = 2/3.

Useful special case: dividing by 3/4 is the same as multiplying by 4/3. This is exactly what you do when you rearrange x = (3/4)y to get y = (4/3)x.

Adding and subtracting fractions

The bottoms must be the same first. Find a **common denominator** (a number both bottoms divide into).

Example: 1/2 + 1/3. Both 2 and 3 go into 6. 1/2 = 3/6 and 1/3 = 2/6. So 3/6 + 2/6 = **5/6**.

Example: 5/2 − 3/4. Common denominator 4. 5/2 = 10/4. 10/4 − 3/4 = **7/4**.

Example: 2 + 3/4. Write 2 = 8/4. 8/4 + 3/4 = **11/4**.

Negative fractions

−(9/5) is the same as (−9)/5 and 9/(−5). The minus can sit anywhere, but only once. Two minus signs cancel: (−3)/(−6) = 3/6 = 1/2. You will use this when working out slopes.

Mini tasks 3

1. Write 5/4 as a decimal.
2. Simplify 12/18.
3. (1/2) × 6 − 3/4
4. 4/3 + 10/3
5. (−8)/(−5) — is it positive or negative?
6. Work out h(x) = (1/2)x − 3/4 when x = 1.

4. Powers, square roots and cube roots

Powers (also called indices or exponents)

A **power** is a short way to write repeated multiplication. 2^5 means 2 × 2 × 2 × 2 × 2 = 32.

- The bottom number (2) is the **base**.
- The small raised number (5) is the **exponent** or **index** (plural **indices**). It says how many times to multiply the base.
- Read 2^5 as "2 to the power 5". Read x^2 as "x **squared**" and x^3 as "x **cubed**".

Two special rules you must know

- Anything to the power 1 is itself: $7^1 = 7$.
- Anything (except 0) to the power 0 is 1: $2^0 = 1$, $10^0 = 1$. This matters a lot for binary in Phase 1.

Powers of 2 — memorise this table

Computers use base 2, so these numbers appear all the time (binary conversion, truth table rows, power sets).

Power	2^0	2^1	2^2	2^3	2^4	2^5	2^6	2^7	2^8	2^9	2^{10}
Value	1	2	4	8	16	32	64	128	256	512	1024

Trick: each value is double the one before it.

Other powers you should know

- Powers of 3: 3, 9, 27, 81, 243.

- Powers of 4: 4, 16, 64, 256, 1024.
- Powers of 10: 10, 100, 1000. The decimal system is built on these.
- Squares: 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144.
- Cubes: $1^3 = 1$, $2^3 = 8$, $3^3 = 27$, $4^3 = 64$, $5^3 = 125$.

Powers of negative numbers

The sign depends on whether the exponent is even or odd.

- $(-2)^2 = (-2) \times (-2) = 4$. Even power gives positive.
- $(-2)^3 = (-2) \times (-2) \times (-2) = -8$. Odd power gives negative.

Careful: -2^2 without brackets means $-(2^2) = -4$. With brackets, $(-2)^2 = 4$.

Square roots

The **square root** of a number asks: "what number, multiplied by itself, gives this?" It is written $\sqrt{}$. So $\sqrt{25} = 5$ because $5 \times 5 = 25$.

- $\sqrt{1} = 1$, $\sqrt{4} = 2$, $\sqrt{9} = 3$, $\sqrt{16} = 4$, $\sqrt{25} = 5$, $\sqrt{81} = 9$.
- Some square roots are not whole numbers: $\sqrt{2} \approx 1.414$, $\sqrt{3} \approx 1.732$, $\sqrt{20} \approx 4.47$. The symbol $\approx$ means "approximately equal to".
- $\sqrt{}$ on its own always means the **positive** root. But the equation $x^2 = 25$ has **two** answers: $x = 5$ and $x = -5$, because $(-5)^2 = 25$ too.
- You cannot take the square root of a negative number using ordinary (real) numbers.

Squaring and square-rooting undo each other

This is the idea behind an **inverse** (Phase 3). If $x \geqslant 0$, then $\sqrt{(x^2)} = x$ and $(\sqrt{x})^2 = x$.

Cube roots

The **cube root**, written $\sqrt[3]{}$, asks: "what number, multiplied by itself three times, gives this?"

- $\sqrt[3]{8} = 2$ because $2 \times 2 \times 2 = 8$.
- $\sqrt[3]{(-8)} = -2$ because $(-2) \times (-2) \times (-2) = -8$. Unlike square roots, cube roots of negatives work.
- So $x^3 = -8$ has exactly **one** answer: $x = -2$. This is your mock Question 10.

Factorial (preview)

n! (read "n factorial") means multiply all whole numbers from n down to 1. 4! = 4 × 3 × 2 × 1 = 24. By definition **0! = 1**. Phase 3 covers this fully.

n	0	1	2	3	4	5
n!	1	1	2	6	24	120

Mini tasks 4

1. 2^6 = ?
2. 5×4^5 = ?
3. $(-3)^2$ and $(-3)^3$
4. Solve $x^2 = 49$ (give all answers).
5. Solve $x^3 = -27$.
6. Is $\sqrt{20}$ bigger or smaller than 4.5?

5. Algebra: letters standing for numbers

The words

- A **variable** is a letter that stands for a number we don't know yet, or a number that can change. Usually x, y, n.
- A **constant** is a fixed number, like 3 or −5.
- A **coefficient** is the number multiplying a variable. In 7x, the coefficient is 7. In x alone, the coefficient is 1.
- A **term** is one piece separated by + or −. In 2x + 3, the terms are 2x and 3.
- An **expression** is a collection of terms with no equals sign, like 2x + 3.
- An **equation** says two things are equal, like 2x + 3 = 11. **Solving** it means finding the value of x that makes it true.
- A **formula** is an equation that links several variables, like y = mx + b.
- **Substitute** means replace a letter with a number. Substituting x = 4 into 2x + 3 gives 2 × 4 + 3 = 11.

Note: 2x means 2 × x, and xy means x × y. The × sign is usually left out.

Collecting like terms

Like terms have the same letter part. 3x and 5x are like terms; 3x and 5 are not.

You can add or subtract like terms: 3x + 5x = 8x. And $x^2 - x - 2 + 2x$ stays as $x^2 + x - 2$ (only the x terms combined).

Solving linear equations: the balance rule

An equation is a balance. **Whatever you do to one side, do the same to the other side.** Your aim is to get x alone.

Undo things in reverse order. The opposite of + is −. The opposite of × is ÷.

Example 1: 2x + 3 = 11

1. Subtract 3 from both sides: 2x = 8.
2. Divide both sides by 2: **x = 4**.
3. Check: 2 × 4 + 3 = 11. Correct.

Example 2: 0 = 3x + 10 — this is how you find where a line crosses the x-axis.

1. Subtract 10: −10 = 3x.
2. Divide by 3: **x = −10/3** ≈ −3.33.

Example 3 (with fractions): (3/4)x − 5/2 = 0

1. Add 5/2 to both sides: (3/4)x = 5/2.
2. Divide by 3/4, which means multiply by 4/3: x = (5/2) × (4/3) = 20/6 = **10/3**.

Example 4 (x on both sides): x + 2 = 3x − 6

1. Subtract x from both sides: 2 = 2x − 6.
2. Add 6: 8 = 2x.
3. Divide by 2: **x = 4**.

Rearranging a formula (making y the subject)

The **subject** of a formula is the letter on its own on one side. In y = 2x + 3, y is the subject. Rearranging uses the same balance rule.

Example 5: Make y the subject of x = 2y + 3. (This exact step is how you find an inverse function in Phase 3.)

1. Subtract 3: x − 3 = 2y.

2. Divide by 2: **$y = (x - 3)/2$**.

Example 6: Make y the subject of $x = y^2 + 2$ (with $y \geqslant 0$).

1. Subtract 2: $x - 2 = y^2$.
2. Square root both sides: **$y = \sqrt{(x - 2)}$**. This is mock Question 21.

Example 7: Make y the subject of $x = (3y - 4)/2$.

1. Multiply by 2: $2x = 3y - 4$.
2. Add 4: $2x + 4 = 3y$.
3. Divide by 3: **$y = (2x + 4)/3$**.

Mini tasks 5

1. Solve $5x - 7 = 18$.
2. Solve $4 = 2x + 10$.
3. Solve $(1/2)x - 3/4 = 0$.
4. Make y the subject of $x = 4y - 5$.
5. Make y the subject of $x = (3/4)y - 5/2$.

6. Expanding, factorising and solving quadratics

You need this for one exam skill: finding where a curve like $x^2 - 4x + 3$ meets a line like $2x - 5$ (Phase 3).

Expanding brackets

Expanding means multiplying out brackets so there are none left.

One bracket: multiply the outside by everything inside. $3(2x - 5) = 6x - 15$.

Two brackets: multiply every term in the first by every term in the second. A memory aid is **FOIL**: First, Outer, Inner, Last.

Example: $(x - 2)(x + 1)$

- First: $x \times x = x^2$
- Outer: $x \times 1 = x$
- Inner: $-2 \times x = -2x$
- Last: $-2 \times 1 = -2$

- Total: $x^2 + x - 2x - 2 = \mathbf{x^2 - x - 2}$

What a quadratic is

A **quadratic** is an expression whose highest power of x is 2. Its general form is $ax^2 + bx + c$, where a, b, c are numbers. In $x^2 - 6x + 8$: $a = 1$, $b = -6$, $c = 8$.

A **quadratic equation** sets it equal to zero: $x^2 - 6x + 8 = 0$. It usually has two solutions, also called **roots**.

Factorising (the reverse of expanding)

Factorising means writing an expression as brackets multiplied together. In this module a is always 1, so the method is simple.

For $x^2 + bx + c$, find **two numbers that multiply to c and add to b**. Call them p and q. Then $x^2 + bx + c = (x + p)(x + q)$.

Example 1: $x^2 - 6x + 8$

- Need: multiply to 8, add to −6.
- Pairs that multiply to 8: 1 and 8, 2 and 4, −1 and −8, −2 and −4.
- Which adds to −6? $-2 + (-4) = -6$. Yes.
- So $x^2 - 6x + 8 = \mathbf{(x - 2)(x - 4)}$.

Example 2: $x^2 - x - 2$

- Need: multiply to −2, add to −1.
- −2 and 1: product −2, sum −1. Yes.
- So $x^2 - x - 2 = \mathbf{(x - 2)(x + 1)}$.

Example 3: $x^2 - 5x - 6$

- Need: multiply to −6, add to −5.
- −6 and 1. So $\mathbf{(x - 6)(x + 1)}$.

Sign tips: if c is positive, both numbers have the same sign as b. If c is negative, the numbers have opposite signs.

Solving: the zero rule

If two things multiply to give 0, at least one of them must be 0. So:

$(x - 2)(x - 4) = 0$ means $x - 2 = 0$ or $x - 4 = 0$, so $\mathbf{x = 2 \text{ or } x = 4}$.

Notice the signs flip: the bracket $(x - 2)$ gives $x = +2$, and $(x + 1)$ gives $x = -1$.

Full method for solving a quadratic

1. Move everything to one side so the other side is 0.
2. Factorise.
3. Set each bracket to 0 and solve.

Worked example: solve $x^2 = x + 2$.

1. Move everything left: $x^2 - x - 2 = 0$.
2. Factorise: $(x - 2)(x + 1) = 0$.
3. **$x = 2$ or $x = -1$**.

If the numbers won't factorise neatly, there is also the **quadratic formula**: $x = (-b \pm \sqrt{(b^2 - 4ac)}) / (2a)$. The ± means "plus or minus", giving two answers. You rarely need it in this module, but it is a safety net.

Mini tasks 6

1. Expand $(x + 3)(x - 5)$.
2. Factorise $x^2 - 5x - 6$.
3. Factorise $x^2 + 7x + 12$.
4. Solve $x^2 - 4x - 5 = x + 1$. (Hint: move everything left first.)

7. The coordinate plane (graphs)

A graph is a flat grid used to draw the relationship between two numbers. You need it for straight lines in Phase 3 (mock Questions 2 and 20).

The parts

- The **x-axis** is the horizontal line (left to right). Numbers increase as you go right; left of the middle they are negative.
- The **y-axis** is the vertical line (bottom to top). Numbers increase as you go up; below the middle they are negative.
- The **origin** is where the two axes cross. It is the point (0, 0).

- A **point** is written as a pair (x, y), called **coordinates**. The first number is always x (across), the second is always y (up or down). Memory aid: "along the corridor, then up the stairs".

Plotting points

- (3, 2): go 3 right, then 2 up.
- (−4, 1): go 4 left, then 1 up.
- (2, −3): go 2 right, then 3 down.
- (0, 5): don't move left or right; go 5 up. This point sits **on the y-axis**.
- (6, 0): go 6 right; don't move up. This point sits **on the x-axis**.

Two facts you will use over and over:

1. Every point on the **y-axis** has x = 0.
2. Every point on the **x-axis** has y = 0.

Order matters

(2, 5) and (5, 2) are different points. A pair where order matters is called an **ordered pair**. You will meet ordered pairs again in sets (Cartesian products) and in functions.

Making a table of values

To draw y = 2x − 1, pick some x values, work out y, then plot each (x, y).

x	−2	−1	0	1	2
y = 2x − 1	−5	−3	−1	1	3

The points (−2, −5), (−1, −3), (0, −1), (1, 1), (2, 3) all lie on one straight line. That line is the **graph** of y = 2x − 1.

Reading a graph without a calculator

In the exam you pick the right graph from four choices. Check two things:

1. **Where does it cross the y-axis?** Put x = 0 into the formula. For y = 2x − 1 that gives y = −1, so the line passes through (0, −1).
2. **Does it go up or down as you move right?** Phase 3 explains this with the slope.

Mini tasks 7

1. Which axis is the point (0, −4) on?
2. Make a table of values for y = −x + 3 using x = −1, 0, 1, 2.
3. Where does y = 3x + 6 cross the y-axis?
4. Is the point (2, 5) on the line y = 2x + 1?

8. Inequality signs

An **inequality** compares two numbers that might not be equal. These signs are essential for intervals (Phase 2) and set builder notation.

Sign	Read as	Example (true)	Includes the number itself?
<	is less than	3 < 5	No — strict
>	is greater than	5 > 3	No — strict
⩽	is less than or equal to	3 ⩽ 3 and 3 ⩽ 5	Yes
⩾	is greater than or equal to	5 ⩾ 5 and 5 ⩾ 3	Yes
≠	is not equal to	4 ≠ 5	—

Memory aid: the open mouth of < and > always faces the **bigger** number. The extra line under ⩽ and ⩾ means "or equal".

Strict versus non-strict

- x < 5 means x can be 4.9, 4.999, anything below 5, but **not** 5 itself.
- x ⩽ 5 means x can be 5 as well.

This difference is exactly the difference between a round bracket ( ) and a square bracket [ ] in an interval. You will see it in Phase 2.

Double inequalities

1 < x ⩽ 100 means **both** 1 < x **and** x ⩽ 100 at the same time. So x is bigger than 1 but at most 100. 1 is not allowed, 100 is allowed.

Negative numbers on the number line

Further left is smaller. So −5 < −2 (−5 is colder than −2). Many students get this wrong.

Solving inequalities

Solve them like equations, with one extra rule: **if you multiply or divide both sides by a negative number, flip the sign.**

- 2x + 1 < 7: subtract 1 to get 2x < 6, then divide by 2 to get x < 3.
- −2x < 6: divide by −2 and flip: x > −3.

Mini tasks 8

1. True or false: −7 > −3.
2. True or false: 4 ⩽ 4.
3. Which whole numbers satisfy 2 < x ⩽ 5?
4. Solve 3x − 4 ⩾ 8.
5. Write "x is at least 2 and less than 9" using signs.

9. Symbol dictionary for the whole module

Maths exams are hard mostly because of the symbols. Here is every symbol in your lectures, how to read it aloud, and a tiny example. Don't memorise it now; come back to it while studying each phase.

Numbers (Phase 1)

Symbol	Read as	Meaning / example
$\mathbb{N}$	N, natural numbers	1, 2, 3, … (Lecture 2 also includes 0)
$\mathbb{Z}$	Z, integers	…, −2, −1, 0, 1, 2, …
$\mathbb{Q}$	Q, rational numbers	Numbers that are fractions of integers, like ½
$\mathbb{R}$	R, real numbers	Every number on the number line, including √2
a mod b	a modulo b	The remainder when a is divided by b. 14 mod 5 = 4
≡	is congruent to	14 ≡ 4 (mod 5): same remainder when divided by 5

Symbol	Read as	Meaning / example
1011_2	1011 in base 2	A small 2 at the bottom means binary
a_n	a sub n	The nth term of a sequence

Sets and intervals (Phase 2)

Symbol	Read as	Meaning / example
{ }	the set of	{1, 2, 3} is a set with three elements
∅ or { }	empty set	A set with nothing in it
∈	is an element of / is in	2 ∈ {1, 2, 3}
∉	is not an element of	5 ∉ {1, 2, 3}
|A|	cardinality of A	How many elements A has. |{a, b}| = 2
⊆	is a subset of	{1, 2} ⊆ {1, 2, 3}
⊂	is a proper subset of	A subset that is not the whole set
∪	union (OR)	{1, 2} ∪ {2, 3} = {1, 2, 3}
∩	intersection (AND)	{1, 2} ∩ {2, 3} = {2}
A \ B	A minus B, set difference	In A but not in B
A' or A^c	complement of A	Everything in the universe U not in A
U	universal set	Everything being considered
P(A)	power set of A	The set of all subsets of A
A × B	Cartesian product	All ordered pairs (a, b)
| or : inside { }	such that	{x | x > 3} = the set of x such that x > 3
(a, b)	open interval	Numbers between a and b, not including a or b
[a, b]	closed interval	Numbers between a and b, including both
∞	infinity	Goes on forever

Functions (Phase 3)

Symbol	Read as	Meaning / example
f(x)	f of x	The output of function f when the input is x
f: A → B	f maps A to B	f takes inputs from A and gives outputs in B
$f^{-1}(x)$	f inverse of x	The function that undoes f. (Not 1/f!)
m	slope	How steep a line is
Δ	delta, change in	Δy = change in y
n!	n factorial	4! = 24
≈	approximately	√2 ≈ 1.414
±	plus or minus	Two answers at once

Logic (Phase 4)

Symbol	Read as	Meaning
T, F	true, false	The only two truth values
¬	not	Flips true and false
∧	and	True only when both sides are true
∨	or	True when at least one side is true
→ or ⇒	implies, if … then	False only when true leads to false

Notice: ∩ (intersection) looks like ∧ (and), and ∪ (union) looks like ∨ (or). That is not an accident. Intersection is "in A **and** in B"; union is "in A **or** in B".

Words, not symbols

- **Iff** means "if and only if".
- **e.g.** means "for example"; **i.e.** means "that is".
- **Respectively** means "in the same order as listed".

Phase 0 practice questions

Do these on paper, without a calculator, before reading the answers. Aim for about 20 minutes. If you get 12 or more right, you are ready for Phase 1.

1. Work out $7 + 2 \times (5 - 8)$.
2. Divide 100 by 7. Give the quotient and the remainder.
3. Work out $3 \times 5 - 2 \times 4 + 2^2$.
4. Work out $(1/2) \times (-3) - 3/4$.
5. Solve $4x - 9 = 2x + 5$.
6. Make y the subject of $x = (y - 3)/5$.
7. Solve $x^2 - 4x + 3 = 2x - 5$. (Move everything to one side first.)
8. Solve $x^3 = 64$.
9. Work out $2^7 + 2^3$.
10. List every integer x with $-2 < x \leqslant 1$.
11. Factorise $x^2 + x - 12$.
12. Where does the line $y = -4x + 8$ cross the y-axis, and where does it cross the x-axis?
13. Multiple choice: which gives every solution of $x^2 = 9$? (A) 3 only (B) −3 only (C) 3 and −3 (D) there are none
14. Work out 5×4^3.
15. Work out $(-1)^2 + (-1)^3 + 0!$.

Answers with full explanations

Mini tasks 1 (numbers)

1. 20 ÷ 6: 6 × 3 = 18 is the biggest multiple of 6 not over 20. Quotient **3**, remainder 20 − 18 = **2**.
2. 23 ÷ 7: 7 × 3 = 21. So **3 remainder 2**.
3. −3 − 8 = **−11** (move 8 further left). −5 + 12 = **7**. 4 − (−6) = 4 + 6 = **10**. (−6) × (−7) = **42** (same signs, plus). (−20) ÷ 5 = **−4** (different signs, minus).
4. 945 ends in 5, so it is **divisible by 5** (945 = 5 × 189). It is odd, so **not divisible by 2**.
5. |−15| = **15**.

Mini tasks 2 (BIDMAS)

1. Division first: 4 ÷ 2 = 2, then 10 − 2 = **8**.
2. Bracket first: 6, then 6 ÷ 2 = **3**.
3. Power first: $3^2 = 9$, then 2 × 9 = **18**.
4. Bracket first: 6, then 6^2 = **36**. Compare with question 3: brackets change everything.
5. Bracket: 2 − 7 = −5. Then 5 − (−5) = 5 + 5 = **10**.

Mini tasks 3 (fractions)

1. 5 ÷ 4 = **1.25**.
2. Divide top and bottom by 6: **2/3**.
3. (1/2) × 6 = 3. Then 3 − 3/4 = 12/4 − 3/4 = **9/4** = 2.25.
4. Same bottoms, add tops: **14/3** ≈ 4.67.
5. Negative divided by negative is **positive**: 8/5 = 1.6.
6. (1/2) × 1 = 1/2 = 2/4. Then 2/4 − 3/4 = **−1/4** = −0.25.

Mini tasks 4 (powers and roots)

1. 2^6 = **64**.
2. $4^5 = 1024$, so 5 × 1024 = **5120**. (This is the revision sheet's geometric sequence answer.)
3. $(-3)^2$ = **9** (even power, positive). $(-3)^3$ = **−27** (odd power, negative).
4. **x = 7 or x = −7**, since both square to 49.
5. **x = −3**, since $(-3)^3 = -27$. Only one answer.
6. $4.5^2 = 20.25$, which is bigger than 20. So $\sqrt{20} \approx 4.47$ is **smaller** than 4.5.

Mini tasks 5 (algebra)

1. Add 7: 5x = 25. Divide by 5: **x = 5**.
2. Subtract 10: −6 = 2x. Divide by 2: **x = −3**.
3. Add 3/4: (1/2)x = 3/4. Multiply by 2: **x = 3/2** = 1.5.
4. Add 5: x + 5 = 4y. Divide by 4: **y = (x + 5)/4**.
5. Add 5/2: x + 5/2 = (3/4)y. Multiply both sides by 4/3: y = (4/3)x + (4/3)(5/2) = **(4/3)x + 10/3**. This is the inverse from revision Task 6.

Mini tasks 6 (quadratics)

1. $x^2 - 5x + 3x - 15 =$ **$x^2 - 2x - 15$**.
2. Multiply to −6, add to −5: −6 and 1. **$(x - 6)(x + 1)$**.
3. Multiply to 12, add to 7: 3 and 4. **$(x + 3)(x + 4)$**.
4. Subtract x and 1 from both sides: $x^2 - 5x - 6 = 0$. Factorise: $(x - 6)(x + 1) = 0$. So **$x = 6$ or $x = -1$**. (This is Seminar 3 Task 5.)

Mini tasks 7 (graphs)

1. $x = 0$, so it is on the **y-axis**.
2. $x = -1$ gives $y = 4$; $x = 0$ gives 3; $x = 1$ gives 2; $x = 2$ gives 1.
3. Put $x = 0$: $y = 6$. It crosses at **(0, 6)**.
4. $2 \times 2 + 1 = 5$. Yes, **(2, 5) is on the line**.

Mini tasks 8 (inequalities)

1. **False.** −7 is further left than −3, so $-7 < -3$.
2. **True.** $\leqslant$ includes equal.
3. **3, 4, 5.** 2 is excluded (strict <), 5 is included ($\leqslant$).
4. Add 4: $3x \geqslant 12$. Divide by 3: **$x \geqslant 4$**.
5. **$2 \leqslant x < 9$**.

Phase 0 practice questions

1. Bracket: $5 - 8 = -3$. Multiply: $2 \times (-3) = -6$. Add: $7 + (-6) =$ **1**.
2. $7 \times 14 = 98$, and $100 - 98 = 2$. Quotient **14**, remainder **2**.
3. Power: $2^2 = 4$. Multiplications: 15 and 8. Then $15 - 8 + 4 =$ **11**.
4. $(1/2) \times (-3) = -3/2 = -6/4$. Then $-6/4 - 3/4 =$ **$-9/4$** $= -2.25$.
5. Subtract 2x: $2x - 9 = 5$. Add 9: $2x = 14$. **$x = 7$**.
6. Multiply by 5: $5x = y - 3$. Add 3: **$y = 5x + 3$**.
7. Subtract 2x and add 5 on both sides: $x^2 - 6x + 8 = 0$. Factorise: $(x - 2)(x - 4) = 0$. **$x = 2$ or $x = 4$**. This is revision Task 8: the curve and the line meet at (2, −1) and (4, 3).
8. $4 \times 4 \times 4 = 64$, so **$x = 4$**.
9. $128 + 8 =$ **136**.
10. −2 is excluded, 1 is included: **−1, 0, 1**.

11. Multiply to −12, add to +1: 4 and −3. **(x + 4)(x − 3)**.
12. y-axis: put x = 0, y = 8, so **(0, 8)**. x-axis: put y = 0, so 0 = −4x + 8, 4x = 8, x = 2, so **(2, 0)**.
13. **(C).** $3^2 = 9$ and $(-3)^2 = 9$. A square has two roots.
14. $4^3 = 64$. 5 × 64 = **320**.
15. $(-1)^2 = 1$, $(-1)^3 = -1$, 0! = 1. So 1 − 1 + 1 = **1**.

When you're done, ask for **Phase 1: Numbers, binary, modular arithmetic and sequences**.